NATIONAL
Junior College

SH2 Preliminary Examinations

8834 H1 Geography

Paper 1

Suggested Marking Guide

Disclaimer:

The Suggested Marking Guide should be treated with confidentiality, and should be used with discretion.

Notes on Marking

Marking Approaches

a. **Point marking** is used for marking all sub-parts of the structured questions, except for the 10-mark fieldwork evaluative sub-part.

For point marking, the general rule is that each creditworthy response is awarded one mark. More detailed guidance is provided in the marking guide for each question.

b. **Generic level descriptors** are used for marking:

- 10-mark fieldwork evaluative sub-part structured questions; and
- 13-mark essays.

Possible approaches are provided for questions marked using generic level descriptors. They provide guidance on how the questions could be approached. The possible approaches provided are **neither exhaustive nor should they be treated as a model approach to questions**.

Generic Level Descriptors

Levels	Marks	Generic Level Descriptors for H1 Fieldwork Evaluative Question
3	9-10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context
2	5-8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1-4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Levels	Marks	Generic Level Descriptors for H1 Essays
4	11-13	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	8-10	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	5-7	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1-4	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Note:
Assessment using generic level descriptors involves qualitative rather than quantitative evaluation. Judgements on the level to be awarded to an answer will be based on the principle of 'best fit' determined by the descriptors within each level.

As a general guideline, responses deemed to have fulfilled all the descriptors within a level may be awarded the top mark in that level.

Section A

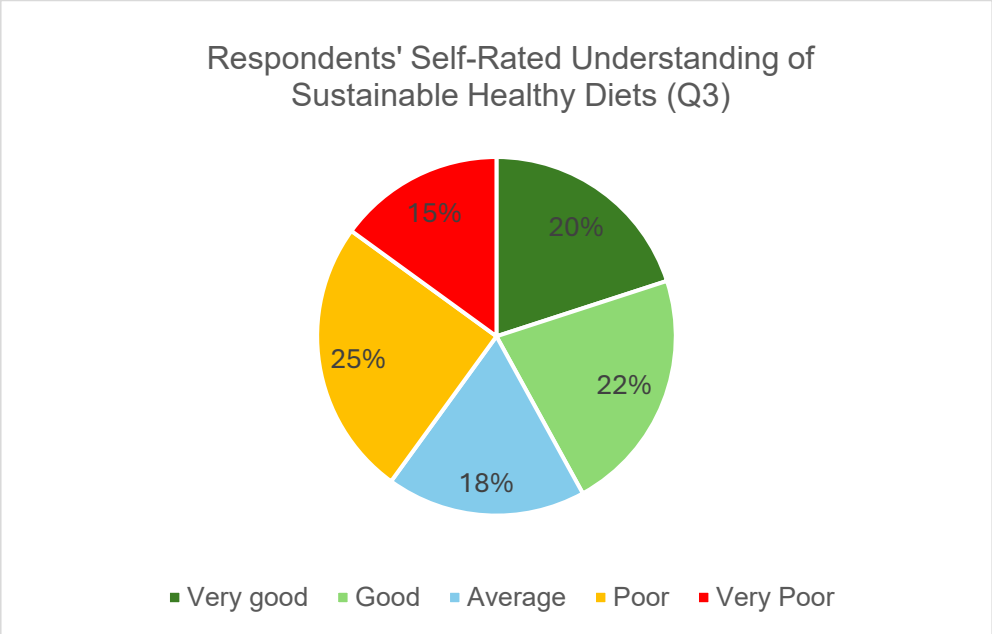
Cluster 2: Fieldwork

Question	Answer	Marks
1	A group of six students in Fremantle, a port city in Australia, wanted to investigate the local community's perceptions about sustainable healthy diet practices. Sustainable healthy diets are defined as "dietary patterns that promote all dimensions of individuals' health and wellbeing; have low environmental pressure and impact; are accessible, affordable, safe and equitable; and are culturally acceptable" (Food and Agriculture Organization of the United Nations (FAO) and World Health Organization (WHO), 2019). They came up with the following hypothesis for their investigation: *Adults living in Fremantle with higher educational attainment (above pre-university level) are more likely to be concerned about sustainable healthy diet practices.* They conducted a survey, using a questionnaire, with 100 residents aged from 18 to 75 at two sites on a Friday from 11 am to 2 pm. They used convenience sampling to select the residents for the survey. Two of the residents also agreed to participate in a face-to-face interview to elaborate on their perceptions of sustainable healthy practices. Resource 1 is a map showing the locations of Site A (Freemantle Markets) and Site B (John Curtin College of the Arts) where the survey was conducted. Resource 2 shows photographs of Sites A and B. Resource 3 shows the number of residents surveyed according to educational attainment. Resource 4 is the questionnaire used in the investigation. Resource 5 shows a summary of responses to selected questions in the questionnaire.	

(a)	**Explain why the students' hypothesis is appropriate for the investigation.** *Award 1 mark for each explanation of why the hypothesis is appropriate, to a maximum of 3 marks.* *Award a maximum of 1 additional mark for further development of each explanation, where applicable.* Possible responses include: • The hypothesis is **clearly defined** with a **specific target group,** i.e. **adults living in Fremantle**, for the investigation. o It also clearly **defines the term "sustainable healthy diet"** by adopting the FAO and WHO's 2019 definition for the investigation. • It is **at a suitable scale** as the investigation is confined **within Fremantle**. **AO1**	**3**
(b)	**With reference to Resources 1, 2 and 3, suggest how the students could have obtained a more representative sample of residents in Fremantle.** *Award 1 mark for each suggestion, to a maximum of 2 marks.* *Award 1 additional mark for further development of each suggestion.* Possible responses include: • Conducting the survey outside Fremantle Markets and John Curtin College of the Arts (Resources 1 & 2) introduces **bias.** o To obtain a more representative sample, the students could have surveyed at **a wider variety of locations across different neighbourhoods** in Fremantle, including residential areas, workplaces and community centres, to capture a broader cross-section of the population. • The students should **reconsider their sampling timing**. Conducting the survey on a Friday between 11 am and 2 pm introduces **bias**. o Surveying during a weekday means the sample likely **over-represents certain groups**, especially those not in full-time weekday employment, while **under-representing** working adults.	**4**

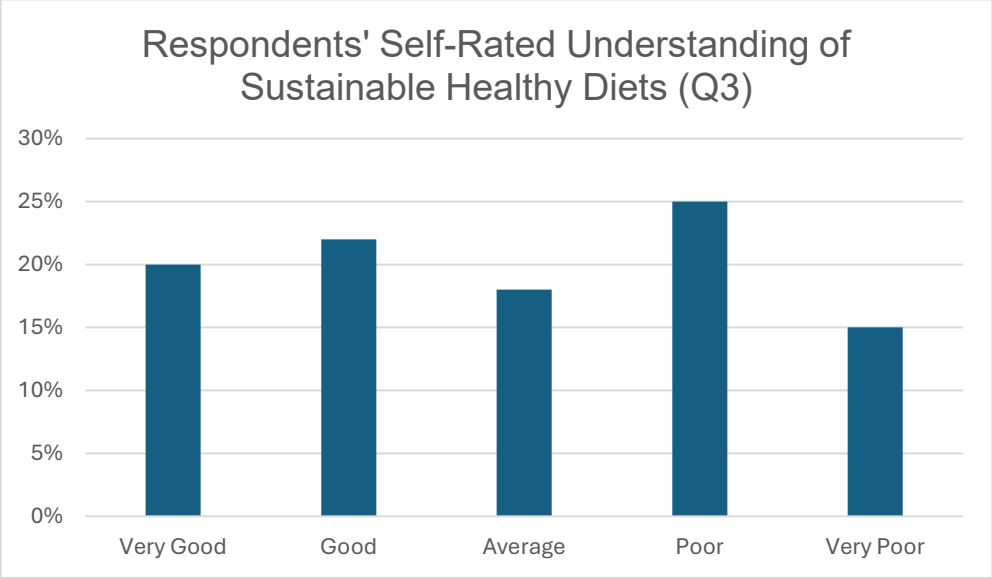
	AO2	
(c)	**With reference to Resource 2, suggest how the students could minimise potential risks when conducting the survey.** *Award 1 mark for each suggestion on how the students could minimise potential risks during data collection, to a maximum of 4 marks.* *Award a maximum of 1 additional mark for further development of each suggestion, where applicable.* Possible responses include: • First, the students should always **work in pairs or trios** rather than alone. This **reduces personal safety risks**, particularly when approaching strangers. o It ensures that there is **always someone nearby in the event of an uncomfortable or threatening interaction** with a member of the public. • Second, the students should **seek the necessary permission / approval** before conducting the survey, i.e. from the management of Fremantle Markets and John Curtin College of the Arts, to ensure they are **legally permitted to conduct surveys on or near those premises**. o This **minimises the risk of being asked to leave or facing legal complications mid-survey**. **AO1**	**4**
(d)	**With reference to Resource 4, explain the strengths and limitations of the questionnaire used in the investigation.** *Award 1 mark for each explanation of strength or limitation, to a maximum of 4 marks.* *Award a maximum of 1 additional mark for further development of each explanation.* *Award a maximum of 2 marks if only strengths or limitations are explained.* *Note: Only about the questionnaire, not about the time or sampling.*	**4**

	Possible responses include: **Strength:** • One strength of the questionnaire is that it is **well-structured and logically organised into two clear sections** – **demographics** as well as **perceptions and attitudes** – making it easy for respondents to follow and complete. o This reduces the likelihood of confusion or incomplete responses, **improving the overall quality of the data collected**. **Limitation:** • One limitation is that several questions are **susceptible to social desirability bias**, where respondents may give answers they feel are socially acceptable rather than truthful. o For instance, in Q6, respondents **may claim to be "very willing" to change their eating habits** to appear environmentally conscious, even if their actual behaviour does not reflect this. This **could lead to an overestimation of concern for sustainable diets** across all educational groups. **AO2**	
(e)	**Sketch a graph to present the data that the students obtained for Q3 in Resource 5 and explain its suitability.** *Award 1 mark each for title, labelled sectors, legend up to a total of 2 marks.* *Award 1 mark for accuracy.* *Award up to 2 marks for justification.* Possible responses include:	**5**



- A **pie chart** is **suitable for displaying percentage data** at a single point in time.
- It allows readers to **quickly and visually compare the relative size of different categories**, making it easy to identify which category is dominant or most significant at a glance.

OR



	• A bar graph enables **easy comparison** of the values across different categories because the **lengths of the bars are directly proportional to the data values**. • A bar graph clearly highlights **the highest and lowest values**, as well as any outliers or significant disparities between categories. **AO2**	
(f)	**Evaluate the validity of the students' investigation.** Possible Approaches: Candidates could evaluate the validity of the investigation by **reviewing the hypothesis and choice of research area / location a** shown in Resources 1 and 2. Candidates could evaluate **the method used to collect the data and the data collected** in Resources 3, 4 and 5. For the data collection method, candidates **could review the questionnaire** to assess the impact on **accuracy and reliability of data collected.** Evaluation could also look into the **surveying method, sample size** and **whether the sample selected is representative** of the adult population in Fremantle. Candidates could consider the **design of the questionnaire,** such as its strengths and limitations and how it would have been **augmented with other relevant data from the in-depth interviews.** Additionally, candidates could consider what other data would be needed for this investigation. For instance, they could mention the need for information such as **collecting observational data** at Fremantle Markets, such as **recording the types of produce purchased** by different shoppers or even getting respondents' **diary records of food purchased over a set period** (e.g. one to two weeks) would provide concrete **behavioural data** to complement the perception / attitudinal data from the questionnaire. Levels marked using modified Generic Level Descriptors for H2 fieldwork evaluative questions.	**10**

	AO3	

Section B

Cluster 1: Sustainable Future and Climate Change

Question	Answer	Marks
2	**Resource 6 shows the location of four waterfront redevelopment projects in Shanghai, China. Resource 7 shows a comparison of the four waterfront redevelopment projects. Resource 8 shows a view of the Pudong financial district in Shanghai in 1987 and 2013.**	
2 (a)	**Using Resources 6 and 7, compare the four waterfront redevelopment projects in Shanghai.** *Award 1 mark for each comparison of the waterfront projects, to a maximum of 4 marks.* *Award 1 additional mark for further development of each comparison.* *Award a maximum of 3 marks for similarities or differences only.* Possible responses include: Similarities: • All four sites share **industrial land as part of their pre-development condition**, with residential land use also present across three of the four sites. • The projects share a broad regenerative intent by **transforming underutilised / derelict waterfront land into a higher-value economic or cultural district**, reflecting a **city-wide strategy** of leveraging waterfront locations for **urban competitiveness**. Differences: • In terms of **scale**, **Lujiazui** is the **largest at 28 sq km**, while the **World Expo Site** is the **smallest at 6.68 sq km**, with Yangpu (12.93 sq km) and Xuhui West Bund (9.4 sq km) falling in between. • Chronologically, **Lujiazui** was the **earliest (1990)**, followed by **Yangpu** and the **World Expo Site (both 2002),** and finally **Xuhui West Bund (2010)**.	**5**

	• The development strategies also differ considerably – **Lujiazui** relied on **special economic zones** to drive financial and commercial growth, the **World Expo Site** used a **mega-event as a catalyst**, **Yangpu** emphasised **entrepreneurship and innovation** and Xuhui West Bund took **a culture-led approach**. **AO2**	
2 (b)	**Suggest reasons for the different development strategies used.** *Award 1 mark for each suggested reason, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggested reason.* Possible responses include: • The differences in development strategies across the four projects can be explained by **a combination of temporal, economic and contextual factors**. • Lujiazui's **special economic zone strategy** reflects the **political and economic climate of the early 1990s**, when China was in the early stages of **economic liberalisation**. Establishing **a new CBD and financial centre** could be a deliberate national priority, and the special economic zone model was the dominant tool available at the time for **attracting foreign direct investment** and **signalling openness to global capital.** • • By 2002, Shanghai had **already established itself as a global financial centre** through Lujiazui, and the focus shifted towards **diversifying the city's economic base**. • The **World Expo Site's mega-event strategy** was **opportunistic** – the hosting of a major international event provided both the **funding justification** and the **global visibility** needed to accelerate redevelopment of a large former industrial area. **AO2**	**4**

2 (c)	**With reference to Resource 8, compare the financial district of Pudong before and after the redevelopment.** *Award 1 mark for each comparison of the financial district of Pudong before and after the redevelopment, to a maximum of 4 marks.* *Award 1 additional mark for further development of each comparison.* *Award a maximum of 2 marks if only similarities or differences are described.* Possible responses include: Similarities: • The **Huangpu River** remains the **defining geographical feature** in both photographs, with **river traffic present in both periods**, suggesting that the **waterfront's role as an active urban corridor** has been maintained. • The **historic area / Bund** characterised by its **colonial-era low-rise buildings** (e.g. the iconic Shanghai Custom House with a clock tower) – remains **largely unchanged**, providing a consistent foreground in both photographs. Differences: • The built environment has been **transformed from an industrial waterfront into a financial district.** o The **most striking change is in the skyline** – the **1987 photo** shows a **flat, low-rise landscape** dominated by **industrial structures, docks and warehouses**, while the **2013 photo** features a **dense cluster of iconic high-rise towers**. **AO2**	**4**
2 (d)	**Account for the varying success of urban reimaging strategies across cities.** *Award 1 mark for each explanation of varying success of urban reimaging strategies, to a maximum of 4 marks.* *Award maximum 1 additional mark for further development of each explanation.* Possible responses include:	**5**

	• The **availability of financial resources** is a key determinant of success. **Wealthier cities or those with strong government backing** can invest more in **large-scale infrastructure, iconic architecture and sustained marketing campaigns**, increasing the likelihood of success. Cities with limited funding may struggle to implement strategies at a scale sufficient to shift perceptions or attract investment. • **Political will and governance capacity** also matter. Cities with **stable, long-term planning frameworks and strong institutional coordination**, such as Singapore or Barcelona, are better placed to implement coherent reimaging strategies. In contrast, cities with fragmented governance or frequent changes in political leadership may lack the consistency needed to see strategies through. • The **pre-existing reputation and global visibility of a city** influences how easily reimaging efforts gain traction. Cities that already have **some degree of international recognition** may find it **easier to reposition their image**, whereas it is more challenging for cities with deeply entrenched negative perceptions to change how they are seen by investors, tourists and residents. • The **choice of strategy relative to local context** also affects success. A strategy that has proven effective in one city cannot necessarily be transplanted to another, as the underlying economy, cultural identity, demographics and physical geography of each city shape whether a strategy is viable. o A **culture-led regeneration approach** is more likely to succeed in a city that has an **established creative sector or a vibrant culture** to build upon but may yield limited results in cities without such foundations. **AO1**	
2 (e)	**Explain the demand placed on natural environments by urban areas.** *Award 1 mark for each explanation of the demand placed on natural environments by urban areas, to a maximum of 4 marks.* *Award maximum 1 additional mark for further development of each explanation.*	**4**

	Possible responses include: • The **physical expansion of cities converts natural land**, such as forests, mangroves and agricultural land, **into built-up areas**, resulting in **habitat loss, reduced biodiversity and the disruption of natural drainage systems**. ○ This urban sprawl also **fragments ecosystems**, limiting the ability of species to move and survive. • Urban areas also place **heavy demands on water resources**, **drawing on rivers, aquifers and reservoirs** for household and industrial uses. This can **deplete freshwater sources** and **alter the hydrology** of surrounding natural environments. • The concentration of economic activity and population in cities generates **large volumes of waste and pollution**, which are often discharged into surrounding natural environments, **degrading air and water quality** and **harming ecosystems beyond the city boundary**. **AO1**	

Question	Answer	Marks
3	**Resource 9 shows the indicators of global climate change over the last 2024 years. Resource 10 shows the simulated annual mean precipitation changes relative to 1850–1900 at global warming level of 2°C.**	
3 (a)	**Using Resource 9, describe the indicators of global climate change over the last 2024 years.** *Award 1 mark for each description of an indicator, to a maximum of 4 marks.* *Award a maximum of 1 additional mark for further development of each description.* Possible responses include: • The graph shows that **all three climate indicators – atmospheric carbon dioxide levels, global temperature change, and the rate of global sea level rise – remained relatively stable for the majority of the 2000-year period**	**4**

	before rising sharply in the modern era, **about 1750,** signified by the invention of steam engine and the Industrial Revolution. • **Carbon dioxide levels** stayed **generally constant at ~275–280 ppm from year 1 CE to ~1750**. o After this point, **concentrations rose steeply and accelerated exponentially, reaching 424 ppm by 2024** – a level far exceeding anything recorded in the preceding two millennia. • **Global temperature change**, measured relative to pre-industrial levels, **fluctuated within a narrow range of roughly 0°C for most of the period**, with natural variability evident in the reconstructed data. o However, **from ~1850 onwards**, observed temperatures **began rising sharply, reaching approximately 1.3°C above pre-industrial levels by 2024**, with the rate of increase steepening in recent decades. • The **rate of global sea level rise similarly remained close to zero or slightly negative till ~1850**, with only very gradual change visible in the reconstructed data. **AO2**	
3 (b)	**Explain why atmospheric carbon dioxide levels are increasing.** *Award 1 mark for each explanation on why carbon dioxide levels are increasing, to a maximum of 4 marks.* *Award maximum 1 additional mark for further development of each explanation.* Possible responses include: • Carbon dioxide levels in the atmosphere are increasing primarily due to **human activities that release stored carbon into the atmosphere at a rate faster than natural systems can absorb it**. • The **burning of fossil fuels**, i.e. coal, oil and natural gas, for **energy production, transportation and industrial processes** is the single largest contributor, releasing carbon that has been locked underground for millions of years.	**4**

	• **Deforestation** aggravates the issue by **removing trees** that would **otherwise absorb carbon dioxide through photosynthesis**, while the **burning of cleared vegetation releases additional carbon** directly into the atmosphere. • **Agricultural practices,** particularly the expansion of **livestock farming and rice cultivation**, also contribute to the release of carbon dioxide. **AO1**	
3 (c)	**Using Resource 10, describe the simulated annual mean precipitation changes.** *Award 1 mark for each description of the simulated annual mean precipitation changes, to a maximum of 4 marks.* *Award a maximum of 1 additional mark for further development of each description.* Possible responses include: • At a global warming level of 2°C, the map shows **a broadly uneven pattern of precipitation change**, with **some regions becoming wetter and others drier** relative to the 1850–1900 baseline. • **Increases in precipitation** are projected across **high latitude regions**, particularly across **northern Canada, northern Europe, northern Asia (Siberia) and Antarctica**, with **increases of around 10 – 30% or more**. • The **equatorial Pacific and parts of North Africa** also show notable increases of **10 – 40%** in precipitation, as do **parts of South and Central Asia of 10 – 30%.** • In contrast, **decreases in precipitation** are projected across **subtropical regions**. The **Mediterranean region, southern Europe and a small part of North Africa** appear particularly affected, with **reductions of around 10 – 20%**. • **Southern Africa, southeastern and southwestern Australia, parts of Central America and the Caribbean** and **a large part of South America** also show **drying trends of generally about 10% with a few areas reaching 20%**. **AO2**	**4**

3 (d)	**Suggest possible reasons for the changes shown in Resource 10.** *Award 1 mark for each suggested reason, to a maximum of 4 marks.* *Award 1 additional mark for further development of each suggested reason.* Possible responses include: • High-latitude regions are projected to receive more precipitation because **warmer air can hold more water vapour** (about 7% more moisture for every 1°C of warming), increasing the amount of moisture available for precipitation. o **Rising temperatures enhance evaporation from oceans, especially in the Arctic and North Atlantic**. • Climate change is expected to **shift storm tracks poleward**, bringing **more frequent and intense storms to higher latitudes**. • Some tropical regions are projected to receive more rainfall due to **higher temperatures intensifying convective rain and leading to stronger rainfall events.** • In the equatorial Pacific, **warmer sea-surface temperatures increase convection**, causing **more moisture-laden air to rise** and produce rainfall. **AO2**	**5**
3 (e)	**Describe how the impact of climate change on aquatic and terrestrial ecosystems may affect humans.** *Award 1 mark for each description, to a maximum of 5 marks.* *Award 1 additional mark for further development of each description.* *Award a maximum of 3 marks if only aquatic or terrestrial ecosystems are described.* Possible responses include: • Climate change affects aquatic ecosystems in ways that have direct consequences for humans. **Rising ocean temperatures and ocean acidification** – caused by the absorption of excess carbon dioxide – **disrupt marine food chains by bleaching coral reefs and reducing fish populations**.	**5**

	○ As **coral reefs support** approximately **a quarter of all marine species**, their degradation **threatens the liveability and food security of coastal communities** that depend on reef fisheries, particularly in tropical regions. • On land, climate change disrupts terrestrial ecosystems in ways that similarly affect human welfare. **Shifts in temperature and precipitation patterns alter the range and seasonal behaviour of species**, including **pollinators such as bees** that are **essential to agricultural productivity**. ○ The decline of pollinator populations **threatens crop yields**, with **implications for global food security**. • **Changes in vegetation cover and soil moisture** also increase the **risk of land degradation and desertification**, reducing the **productive capacity of agricultural land** and **displacing rural communities**. **AO1**	

Section C

Cluster 1: Sustainable Future and Climate Change

Question	Answer	Marks
4	'Efforts to enhance liveability always contribute to sustainability in cities.' Evaluate this statement.	**13**
	Possible Approaches: Candidates could approach the question by making a judgement on **whether efforts to enhance liveability *always, sometimes or rarely* contribute to sustainability**, through a consideration of the **relationship between liveability and sustainability as concepts** and the extent to which strategies aimed at improving one necessarily advance the other. They could structure their evaluation by examining specific liveability strategies and assessing their sustainability outcomes across environmental, economic and social dimensions.	

	Candidates could also identify **tensions and trade-offs where liveability improvements come at a cost to sustainability**. For example, the construction of large-scale green spaces or recreational infrastructure may enhance residents' quality of life but involve significant resource consumption, land clearance or carbon-intensive construction processes. Similarly, urban regeneration efforts that attract high-income residents may improve perceived liveability while displacing lower-income communities, undermining social sustainability. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	
5	Evaluate the extent to which urban planning has succeeded in creating more liveable cities for women.	**13**
	Possible Approaches: Candidates could approach the question by making a judgement on the **degree of success of urban planning and policy in improving liveability for women**, arguing that **success has been partial, uneven, or context-dependent rather than absolute**. They could examine specific interventions – such as gender-sensitive public space design, women-only public transport carriages, urban lighting programmes and the provision of childcare facilities – and assess how effectively these have addressed women's distinct liveability needs around safety, mobility, housing, and economic participation. Candidates could also **evaluate the limitations of planning and policy** in this regard. They could argue that **many interventions address only the visible, spatial dimensions of liveability** – such as safety in public spaces – while **failing to tackle the structural inequalities** that constrain women's lives more broadly, such as unequal caregiving burdens, limited affordable housing and underrepresentation in planning and policymaking roles. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	

6	'Not all countries have equal capacity to respond effectively to climate change.' Evaluate this statement.	**13**
	Possible Approaches: Candidates could approach the question by making a judgement on the **extent to which response capacity to climate change varies across places**, through a consideration of **factors such as levels of economic development, technological expertise, institutional capacity and political will**. They could argue that CHLDs generally possess greater adaptive capacity – for instance, through the ability to invest in sea walls, early warning systems, drought-resistant crops, and climate-resilient infrastructure – while CLLDs and small island states face significant constraints in mounting effective responses despite often bearing a disproportionate share of climate impacts. Candidates could also evaluate the extent to which **response capacity varies not only between nations but also within them**, considering how factors such as income inequality, age, gender and marginalisation influence the ability of different groups to respond effectively to climate change. They could further consider **whether international mechanisms** – such as the Green Climate Fund, technology transfer agreements under the UNFCCC – **have succeeded in narrowing the gap in response capacity between places**, or whether structural inequalities continue to limit their effectiveness. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	